

EDA Summary: On-street Car Parking Sensor Data (2012)

Monday, 10 August 2026 Sprint 1 – EDA Summary

Team: Team B Dataset: On-street Car Parking Sensor Data – 2012

Research Question: How does parking use change after bike lanes are constructed?

Overview

This EDA covers the 2012 annual on-street car parking sensor file. The full file contains approximately 52.5 million rows and 14 columns. Each row represents a single vehicle arrival/departure event at a sensor, not a static location record. the schema and quality patterns are consistent across parts.

This is a complete calendar-year file and forms a strong pre-intervention baseline for the majority of protected bike lane and pedestrianisation projects in the Infrastructure Victoria study.

Column Summary

Column	Description	Nulls (observed)
DeviceId	Sensor hardware identifier	0%
ArrivalTime	Event timestamps, format MM/DD/YYYY	0%
DepartureTime	HH:MM:SS AM/PM	
DurationSeconds	Length of the parking event in seconds	0%
StreetMarker	Bay-level identifier (e.g. C3176, 1583S)	0.05%
Sign	Parking restriction signage text for the bay	0.02%
Area	Named precinct (e.g. County, Princes Theatre, Southbank, Victoria Market)	0.02%
StreetId	Street-level identifier	0.02%
StreetName	Street name - near 1:1 with StreetId	0.02%
BetweenStreet1 BetweenStreet2	Cross-streets bounding the segment	0.02%

Column	Description	Nulls (observed)
Side Of Street	Side-of-street code (1–5)	0.02%
In Violation	Whether the parking event breached signed restrictions	0%
Vehicle Present	Occupancy flag for the event	0%

Data Quality Findings

- Very low missingness. dimension fields (DeviceId, ArrivalTime, DepartureTime, DurationSeconds, In Violation, Vehicle Present) show effectively zero nulls once the file is properly parsed. Location-related fields also have very low missing values.
- Thousand separator commas found in DeviceId and DurationSeconds (for example "2,045", "1,174"). These commas will be removed before numeric conversion.
- “OLD” token in Sign. A minority of records contain the word “OLD” inside the Sign field.
- Very short parking stays found in DurationSeconds. Short numbers of 5–30 second events are present (likely to be the sensor noise).
- Very long parking stays found in DurationSeconds. Some events exceed 12 hours (overnight or all-day parking).
- StreetId / StreetName reliability. StreetName is a clean, usable street-level key and is the primary field for matching to the IV treatment/control list (after standardisation). StreetId shows a near 1:1 relationship with StreetName in the data.
- Most events are heavily concentrated in the City of Melbourne CBD and inner suburb areas (Lonsdale, Collins, Bourke, Flinders, Queen, Elizabeth, Russell, William, King, Spencer, La Trobe, Southbank, Queensberry / Victoria Market, Jolimont, etc.). This coverage aligns well with most IV treatment and control streets provided by PO.

Join Compatibility Assessment

- StreetName is the strongest practical join field for matching to the IV treatment/control street list. But standardisation is required (for example. “LA TROBE STREET” and “La Trobe St”).
- StreetId is a clean internal street level key (near 1:1 with StreetName) and should be tested against street_spatial.gpkg / IV segment identifiers.
- StreetMarker is the natural bay-level key, comparable to marker_id / KerbsideID in the static parking bays dataset.
- DeviceId is usable within 2012 but should not be assumed stable across years until the reuse check is confirmed on the full file.
- KerbsideID-style identifiers from the static bays file remain unreliable (high missingness + reuse) and should not be the primary bridge.

Recommended matching sequence:

1. Standardise StreetName.
2. Filter events to IV treatment and control streets.
3. Spatially validate against street_spatial.gpkg where needed.

Suitability for Before/After Comparison

This dataset is well suited to the study’s core comparison. It is genuine event-level occupancy data with full-year coverage and precise timestamps, where the On-street Parking Bays dataset, which is a single static snapshot.

Strengths

- Clean 2012 pre-intervention baseline for almost all IV treatment streets.
- Supports calculation of average/median duration, violation rate, approximate occupancy (from event durations), turnover, and peak-hour patterns.
- StreetName allows matching to the IV treatment/control list.

Limitations

- True continuous occupancy is not directly recorded and must be approximated from event durations.
- Later years (post-intervention) are required for a full Difference-in-Differences design.
- DeviceId stability across years still needs confirmation.

Overall assessment Highly suitable as the pre-intervention baseline layer. When combined with equivalent files for later years and the IV treatment/control street list, it can directly support a rigorous before/after or Difference-in-Differences analysis of parking utilisation.

Next Steps

1. Capture the full DurationSeconds distribution and decide on filter out the noise, for example: very short, very long stays.
2. Extract and standardise the exact list of IV treatment and control street names, then filter the 2012 data to those streets.
3. Produce a 2012 baseline comparison (treatment vs control) on key metrics.
4. Repeat this profile for additional years (2013–2020) as they become available to build the full before/after time series.